

Iron is not responsible for *Didymosphenia geminata* bloom formation in phosphorus-poor rivers

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Abstract: Blooms of the river benthic diatom *Didymosphenia geminata* are an enigma because they occur under phosphorus-poor conditions. A recent proposal that ferric–ferrous iron redox shifts sequester the additional phosphorus needed to stimulate and sustain *D. geminata* blooms does not agree with published experimental data showing that blooms only occur when cells are phosphorus-limited. The “iron hypothesis” also infers that blooms would be favoured in rivers with elevated iron, and management should target iron. Surveys of rivers around the world affected by *D. geminata* show that blooms most often occur in iron-poor rivers. Phosphorus uptake experiments conducted under realistic environmental conditions with living *D. geminata* colonies showed no effect of iron enrichment on phosphorus uptake. Iron does not solve the mystery of *D. geminata* growth causing nuisance blooms worldwide.

Résumé : Les proliférations de la diatomée benthique de rivière *Didymosphenia geminata* constituent une énigme puisqu'elles se produisent dans des conditions d'appauvrissement en phosphore. Il a récemment été proposé que des changements d'oxydoréduction du fer ferrique au fer ferreux retiennent le phosphore supplémentaire nécessaire pour stimuler et soutenir les proliférations de *D. geminata*. Ce postulat ne concorde toutefois pas avec les données expérimentales publiées qui démontrent que les proliférations ne se produisent que quand le phosphore est restreint. Il décelle également de cette « hypothèse du fer » que les rivières présentant de fortes teneurs en fer favoriseraient ces proliférations et que les mesures d'aménagement devraient cibler le fer. Les études de rivières affectées par *D. geminata* aux quatre coins du monde montrent que les proliférations se produisent le plus souvent dans des rivières pauvres en fer. Des expériences de mobilisation du phosphore menées dans des conditions ambiantes réalistes avec des colonies de *D. geminata* vivantes n'ont montré aucun effet d'un enrichissement en fer sur la mobilisation du phosphore. Le fer ne résout pas l'énigme des formes de croissance de *D. geminata* qui entraînent des proliférations néfastes partout sur la planète.

[Traduit par la Rédaction]

Sundareshwar et al. (2011) have proposed that complexes between iron (Fe) and phosphorus (P) that occur on the polysaccharide stalks of the benthic diatom *Didymosphenia geminata* play a central role in the development and maintenance of blooms in oligotrophic streams. In this view, ferric–ferrous

redox shifts within mats result in P accumulation and recycling, providing *D. geminata* with an advantage not available to competing species. The importance of ferric–ferrous Fe redox shifts in the cycling of P in lakes has been known since the 1940s (Mortimer 1941) and is one of the most important

Received 6 July 2012. Accepted 11 September 2012. Published at www.nrcresearchpress.com/cjfas on 19 October 2012. J2012-0309

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biogeochemical processes influencing productivity of many freshwater lakes and some streams. However, an association between Fe and *D. geminata* dominance of river benthic communities is not supported by water chemistry data from multiple regions experiencing *D. geminata* blooms, and survey data show that elevated P concentrations that can occur in interstitial waters of *D. geminata* mats do not influence the rate of division of *D. geminata* cells on the periphery of colonies. Further, sorption of Fe and other metallic cations onto algal exopolysaccharide is well known and often occurs in high P environments (Lawrence et al. 1998). Here we outline evidence that Fe–P complexes that occur on *D. geminata* stalks are not responsible for the development or the maintenance of blooms in oligotrophic rivers.

Published research from New Zealand

Experiments in New Zealand show that *D. geminata* maintains its dominance of river benthic algal communities while remaining in a highly P-limited state (Bothwell and Kilroy 2011; Kilroy and Bothwell 2012). P-limited cell division rates of *D. geminata* were not influenced by attachment to thick mats of polysaccharide stalk material (Bothwell and Kilroy 2011). Experiments document that the elongation of *D. geminata* stalks, associated with blooms, preferentially occurs under P-limited conditions (Kilroy and Bothwell 2011), and blooms did not form when supplemental P was available (Kilroy and Bothwell 2012). Surveys around the world find that *D. geminata* blooms are usually associated with very low P conditions (Bothwell et al. 2009). While this could be interpreted as an indication that *D. geminata* has access to P sources not available to other taxa, the application of the frequency of dividing cells technique to *D. geminata* in situ across a spectrum of dissolved reactive phosphorus (DRP) concentrations in rivers on South Island, New Zealand, indicated that this was not the case (Kilroy and Bothwell 2012). Kilroy and Bothwell (2012) found that cell division rates of *D. geminata* correlated positively with DRP in the overlying water and that bloom formation was inversely related to DRP. *Didymosphenia geminata* blooms end when cell division rates are not limited by P, and blooms do not begin unless these rates are P-limited.

Fe concentrations in rivers with and without *D. geminata* blooms

In natural oxygenated waters, Fe exists primarily as a finely divided precipitate of hydrated ferric hydroxide, with particle sizes ranging from filterable ($>0.45\ \mu\text{m}$) to colloidal ($0.001\text{--}0.5\ \mu\text{m}$) (Wetzel 2001). For this reason the fraction of total iron (Fe-T) operationally defined by filtration as dissolved (Fe-D) varies widely depending on the size distribution of ferric hydroxide particles and the porosity characteristics of filters. In the case of rivers affected by *D. geminata* on Vancouver Island, British Columbia, the percentage of Fe-T classified as Fe-D spans the range from 3.5% to 100%, with a median of 44% ($n = 216$; Supplemental Table S1¹). Because of this wide variability, caution is needed when comparing Fe data. Unfortunately, in supporting their contention that higher Fe levels in rivers might foster *D. geminata* blooms, Sundareshwar et al. (2011) compared Fe-T in affected rivers, with Fe-D in non-*D. geminata*

rivers (see supplemental table S1 in Sundareshwar et al. 2011). Our comprehensive surveys show no consistent pattern between Fe-T or Fe-D in river water and the blooming behaviour of this nuisance alga. Moreover, it is unclear what, if any, temporal changes in Fe or other ions (e.g., sulphate) have occurred at the disparate sites where *D. geminata* is blooming.

Many streams in the Black Hills, South Dakota, have naturally high levels of Fe, and in those with highest Fe, ferric hydroxide precipitate significantly reduces attached algal biomass (Holcomb 2002). Fe-T levels in Rapid Creek (mean $0.31\ \text{mg}\cdot\text{L}^{-1}$, $n = 3$ (Sundareshwar et al. 2011) versus mean $0.14\ \text{mg}\cdot\text{L}^{-1}$, median $0.079\ \text{mg}\cdot\text{L}^{-1}$, $n = 64$ (our 2008–2009 surveys); Table 1a, Supplemental Table S2¹), the stream affected by *D. geminata* that was the focus of the Sundareshwar et al. (2011) study, are at the lower end of the range for both mean and median Fe-T values for Black Hills streams (Supplemental Table S2¹). Rapid Creek remains the only stream in the region affected by extensive *D. geminata* blooms.

The absence of association between *D. geminata* colonies and Fe-T was also evident within Rapid Creek. Although *D. geminata* was widespread within the stream, colonies were not visible at all sites or on all occasions. The average Fe-T concentration at sites and on occasions with colonies was not different from that when colonies were absent (Supplemental Table S2¹). Likewise, *D. geminata* standing crop in Rapid Creek, quantified using a biovolume index, was unrelated to Fe-T concentration of the water (Supplemental Fig. S1¹). Collectively, these results indicate no positive relationship between the concentration of Fe-T in river water and *D. geminata* colonies within or among streams in the Black Hills of South Dakota.

Didymosphenia geminata blooms commencing in 2006 in Quebec spread to numerous rivers along the Gaspé Peninsula. Those rivers eventually supporting blooms had significantly lower levels of Fe-D than those rivers remaining *D. geminata* bloom free (Table 1b; Supplemental Table S3¹). In Quebec rivers, alkalinity and pH are related to the distribution of *D. geminata* blooms (Gillis et al. 2010).

Seven rivers on Vancouver Island supporting large *D. geminata* blooms had median Fe-D and Fe-T concentrations of 0.037 and $0.08\ \text{mg}\cdot\text{L}^{-1}$, respectively (Table 1c). Mean Fe concentrations for these rivers ranged between 0.028 and $0.058\ \text{mg}\cdot\text{L}^{-1}$ for Fe-D and between 0.08 and $0.17\ \text{mg}\cdot\text{L}^{-1}$ for Fe-T (Supplemental Table S1¹). Fe-T data were available in two rivers both prior to and during *D. geminata* bloom years. Lower Fe-T during blooms in the Puntledge River might indicate adsorption onto stalk material. However, such a decline was not observed in the Oyster River (Table 1c).

Two remote, adjacent Canadian Rocky Mountain streams were monitored for 8 years as part of a long-term watershed study. *Didymosphenia geminata* was not detected in either stream until 2010, when blooms commenced in one (North York). Both streams had the same level of Fe-T for the entire period of study (Table 1d). Concentrations of Fe-T (median $0.01\ \text{mg}\cdot\text{L}^{-1}$) in these high elevation streams were among the lowest in our surveys and did not differ significantly between the streams affected and unaffected by *D. geminata* (Table 1d).

On South Island, New Zealand, Fe-D data were available from several rivers affected by *D. geminata* and their spring-

¹Supplementary data are available with the article through the journal Web site at <http://nrcresearchpress.com/doi/suppl/10.1139/f2012-112>.

Table 1. Summary of iron (Fe) and dissolved organic carbon (DOC) concentrations measured in rivers in Canada, USA and New Zealand, in relation to the occurrence of *D. geminata* blooms.

	No. of rivers or sites	Total Fe (mg·L ⁻¹)				Dissolved Fe (mg·L ⁻¹)				DOC (mg·L ⁻¹)			
		Sample				Sample				Sample			
		<i>n</i>	Med.	Max.	Statistical test	<i>n</i>	Med.	Max.	Statistical test	<i>n</i>	Med.	Max.	Statistical test
(a) Black Hills, South Dakota													
<i>D. geminata</i> affected	1	64	0.079	1.57	Mann–Whitney <i>U</i> test; <i>p</i> = 0.028		NA			NA			
Not affected	6	57	0.15	8.01									
(b) Quebec													
<i>D. geminata</i> affected	12		NA	NA		576	0.035	1.60	Two-tailed <i>t</i> test; <i>t</i> = 6.51, <i>p</i> < 0.001	511	2.4	50.0	Two-tailed <i>t</i> test; <i>t</i> = 6.17, <i>p</i> < 0.001
Not affected	17		NA	NA		510	0.150	4.13		501	4.8	94.0	
(c) Vancouver Island													
<i>D. geminata</i> affected	7	567	0.08	2.39		271	0.037	0.370		NA	NA	NA	
Before <i>D. geminata</i>	Oyster	104	0.1	1.96							NA	NA	
After <i>D. geminata</i>		24	0.1	0.43	Mann–Whitney <i>U</i> test; <i>p</i> = 0.460					NA	NA	NA	
Before <i>D. geminata</i>	Puntledge	19	0.09	0.96	Mann–Whitney <i>U</i> test; <i>p</i> < 0.005					NA	NA	NA	
After <i>D. geminata</i>		32	0.041	0.335						NA	NA	NA	
(d) Canadian Rockies													
<i>D. geminata</i> affected	North York	118	0.01	0.98	Nonparametric sign test; <i>p</i> < 0.461	NA	NA	NA		137	1.05	6.1	Nonparametric sign test; <i>p</i> < 0.001
Not affected	Star	118	0.01	0.36		NA	NA	NA		158	1.3	7.9	
(e) South Island, New Zealand													
<i>D. geminata</i> affected	10	NA	NA	NA		21	0.016	0.071	Mann–Whitney <i>U</i> test; <i>p</i> = 0.463	Note: <i>g</i> ₄₄₀ values available; see Supplemental Fig. S2			
Not affected	25	NA	NA	NA		34	0.011	0.67					
(f) Colorado Rocky Mountains													
<i>D. geminata</i> affected	6	24	0.079	0.315	Two-tailed <i>t</i> test; <i>t</i> = 3.35, <i>p</i> < 0.009	24	0.036	0.196	Two-tailed <i>t</i> test; <i>t</i> = 3.12, <i>p</i> < 0.012	24	6.2	10.9	Two-tailed <i>t</i> test; <i>t</i> = 2.13, <i>p</i> < 0.07
Not affected	5	20	0.017	0.079		20	0.004	0.028		20	2.6	5.9	

Note: Comparisons are between medians or single values from multiple rivers or sites. NA refers to no data available or not applicable. Statistical tests used for the comparisons were selected according to the data distribution and type (paired or independent). Complete data sets are in Supplemental Tables S1–S5¹.

fed tributaries in 2007 and in 2011 from 19 South Island river sites selected randomly from the New Zealand National Water Quality Network (Supplemental Table S4¹). Fe-D spanned a wide range, and there was no significant difference between Fe-D concentrations in rivers affected or unaffected by *D. geminata* (Table 1e). None of the sites with blooms had Fe-D > 0.071 mg·L⁻¹.

The only data set in which *D. geminata* bloom formation was associated with higher levels of Fe was from a suite of 11 Rocky Mountain streams near Gunnison, Colorado. Six streams with *D. geminata* blooms in 2010 had significantly higher levels of both Fe-D and Fe-T compared with five other nearby streams without *D. geminata* blooms during the same period (Table 1f; Supplemental Table S5¹). Experiments were undertaken to determine if higher Fe concentrations might stimulate P uptake by *D. geminata* colonies.

Fe enrichment experiments in the Colorado Rocky Mountains

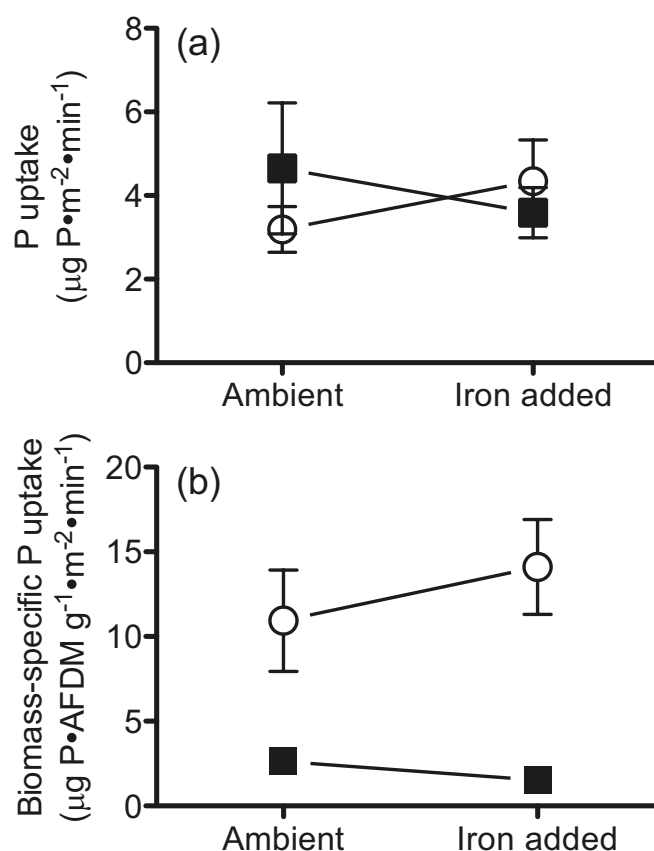
Sundareshwar et al. (2011) used poisoned, nonliving segments of *D. geminata* colonies to demonstrate that additions of Fe increase sequestration of P onto *D. geminata* stalks. They also used concentrations of Fe and P in their trials that were three to six orders of magnitude greater than dissolved levels in typical rivers dominated by *D. geminata*. Our P-uptake experiments with Fe enrichment were conducted with live material on substrates from Rocky Mountain streams incubated in containers for 2 h under in situ stream conditions with 20 µg P·L⁻¹, sodium phosphate added, and with or without Fe additions of 0.3 mg Fe·L⁻¹.

Our realistic level of Fe augmentation failed to increase the uptake of P, and rates of P uptake were similar between substrates with versus without visible colonies of *D. geminata* (Fig. 1a). Biomass-specific P uptake was five times higher on substrates without *D. geminata* colonies than on substrates with colonies and was also not affected by the addition of Fe (Fig. 1b). There was also no interaction between Fe and *D. geminata* colonies on P uptake (Fig. 1b). Hence, using live material under environmentally realistic conditions, the uptake of P by *D. geminata* was not stimulated by additional Fe. More importantly, P removal from the water column (adsorption or uptake) by substrates with *D. geminata* colonies was significantly lower on a per ash-free dry mass basis than P removal by substrates without *D. geminata* colonies, indicating the importance of cellular uptake rather than noncellular adsorption onto *D. geminata* polysaccharide stalks as has been proposed. In addition, these experimental results also suggest that the low P concentration typical of streams in which *D. geminata* blooms (Kilroy and Bothwell 2012) is not a result of greater uptake.

Reactivity of Fe

Fe–organic complexes are important in increasing Fe levels in surface waters but are problematic in predicting Fe reactivity because dissolved organic carbon (DOC) in natural waters is an unknown mixture of compounds with binding constants for Fe that are either unknown or unpredictable among natural waters (Schecher and McAvoy 2003). The best that can be done to evaluate the potential for DOC to hinder Fe–P complexes is to compare DOC concentrations in rivers.

Fig. 1. Effects of *D. geminata* and iron on water column phosphorus. (a) Removal rates (± 1 standard error of the mean, SE) of phosphorus (20 µg P·L⁻¹, as sodium phosphate) by substrates covered with visible colonies of *D. geminata* (solid squares) versus substrates lacking visible colonies of *D. geminata* (open circles) at ambient iron concentrations and with iron added (0.3 mg Fe·L⁻¹, as ferrous sulfate) did not differ (two-way analysis of variance (ANOVA), $F_{[3,31]} = 1.65$, $p = 0.19$). (b) Phosphorus removal rates (± 1 SE) normalized to ash-free dry mass (AFDM) on the substrates were significantly higher in the absence of *D. geminata* colonies (two-way ANOVA, $F_{[1,31]} = 38.63$, $p < 0.0001$) but also not affected by the addition of iron (two-way ANOVA, $F_{[1,31]} = 0.031$, $p < 0.86$). Uncovered containers with substrates from Washington Gulch and Coal Creek were incubated at ambient stream temperature and light.



Across a broad range, rivers dominated by *D. geminata* tend toward lower DOC (Bothwell et al. 2009). It is unclear if the negative relationship between *D. geminata* blooms and DOC is due to reduced light penetration in highly coloured rivers or the association with lower pH in more humate-stained waters. In the view of Sundareshwar et al. (2011), low DOC might also favour *D. geminata* blooms if DOC chelation of Fe reduces reactivity and minimizes sequestration of P. However, we found little evidence to support the latter argument.

In the Colorado Rocky Mountain streams where *D. geminata* blooms were associated with higher concentrations of Fe-T and Fe-D, DOC was also twice as high, potentially off-setting the higher Fe (Table 1f). Median DOC concentrations in two high elevation streams in the Canadian Rocky

Mountains were the lowest of all streams surveyed (1.1 and 1.3 mg·L⁻¹; Table 1d), and although the mean DOC was higher in the stream free of *D. geminata* compared with the stream affected by *D. geminata*, the differences were small (<0.5 mg·L⁻¹; Table 1d). In New Zealand, there was no relationship between the coloured DOC content of river water and the presence or absence of *D. geminata* blooms (Supplemental Fig. S2¹). Only the Quebec data set showed lower concentrations of DOC in rivers affected by *D. geminata*, but they were also associated with Fe-D concentrations four times lower than in rivers lacking blooms (Table 1b).

Conclusion

Values of Fe-T in natural surface waters typically range between 0.05 and 0.20 mg·L⁻¹ (Wetzel 2001). We have found that *D. geminata* often blooms in rivers at the lower end of this spectrum, which contradicts the inference of Sundareshwar et al. (2011) that higher Fe concentrations favour *D. geminata* blooms. More importantly, quantitative measurements indicate that *D. geminata* blooms do not rely on P sourced from within the mat, and uptake rates of P by living *D. geminata* colonies are not increased by realistic levels of Fe enrichment. Rather, *D. geminata* blooms occur and are sustained because of an inadequate supply of P. This conclusion is consistent with observations that *D. geminata* blooms disappear in river reaches immediately downstream of point-source nutrient outfalls in Canadian and US Rocky Mountain rivers (M.L. Bothwell and B.W. Taylor, personal observation) and observations in South Dakota that nutrients added to a stream curtailed *D. geminata* bloom coverage up to 0.6 km downstream (James and Chipps 2012).

The global emergence and spread of *D. geminata* is widely attributed to new introductions. While the root causes of blooms in Northern Hemisphere environments where *D. geminata* is thought to be endemic are unknown, the association between *D. geminata* bloom formation and very low P suggests increasing intensity and frequency of P limitation in pristine waters, perhaps associated with global nitrogen deposition or other regional mechanisms of P depletion. While this remains speculation, the available data do not support the idea that Fe is responsible for promoting and sustaining *D. geminata* blooms in P-poor rivers.

Acknowledgements

Steven R. Chipps (United States Geological Survey, South Dakota State University) provided some of the Fe data from streams in the Black Hills of South Dakota. Susie Wood (Cawthron Institute) coordinated the Fe analyses for New Zealand rivers in 2011. John Deniseger (BC Ministry of Environment) assisted with the compilation of Fe data from the British Columbia Environmental Monitoring System database. This research was supported by Dartmouth College funds to BWT and a National Science Foundation Research Experience for Undergraduates to ETE through Rocky Mountain Biological Laboratory.

P. Chambers, Y. deLafontaine, S. Chipps, and G. Scrimgeour pre-viewed the manuscript.

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